

EXPERIMENT NO. 05

TITLE: FM DEMODULATION



EC39001: Analog Communication Lab
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Theory:

- Frequency Modulation (FM) is a modulation technique where the **frequency of the carrier** is varied according to the amplitude of the message signal.
- Demodulation means recovering the original message signal from the FM wave.
- A **Phase-Locked Loop (PLL)** is a feedback control system that automatically synchronizes the frequency and phase of a Voltage-Controlled Oscillator (VCO) with an input signal.
- By using this property, a PLL can be employed as an **FM demodulator**.

Principle of Operation

1. PLL Basics

A PLL consists of three main components:

- **Phase Detector (PD)** – compares the phase of input FM signal and VCO output.
- **Low-Pass Filter (LPF)** – removes high-frequency components, leaving the error signal.
- **Voltage Controlled Oscillator (VCO)** – generates a signal whose frequency depends on input voltage.

2. Locking Condition

- When the PLL is in lock, the VCO frequency follows the instantaneous frequency of the FM signal.
- The output of the **phase detector + filter** (i.e., control voltage to the VCO) represents the difference between the instantaneous frequency of the FM signal and the VCO's natural frequency.

3. Message Signal Recovery

- In FM, the instantaneous frequency deviation $\Delta f(t)$ is proportional to the message signal $m(t)$.
- Thus, the control voltage of the VCO also becomes proportional to $m(t)$.
- This control voltage is taken as the **demodulated output**.

Mathematical Explanation

FM Wave:

$$S(t) = A_c \cos \left(\omega_c t + K_f \int m(t) dt \right)$$

Where K_f = frequency sensitivity.

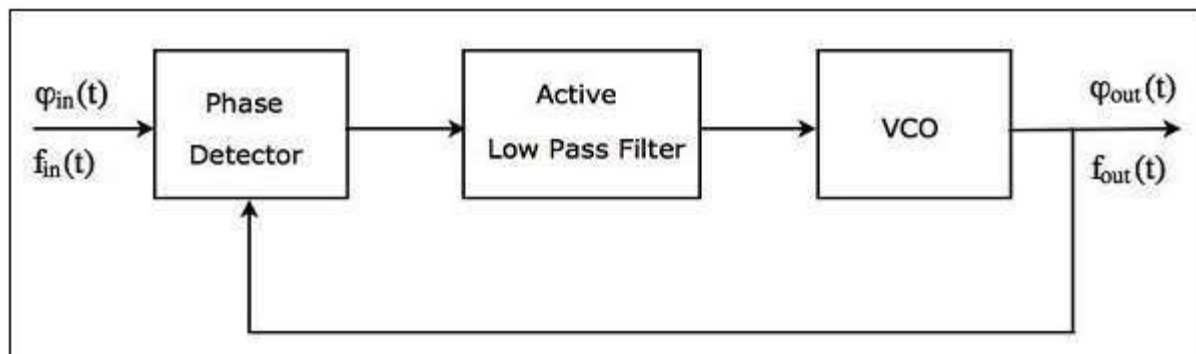
VCO Output:

Phase Detector Output (Error Signal):

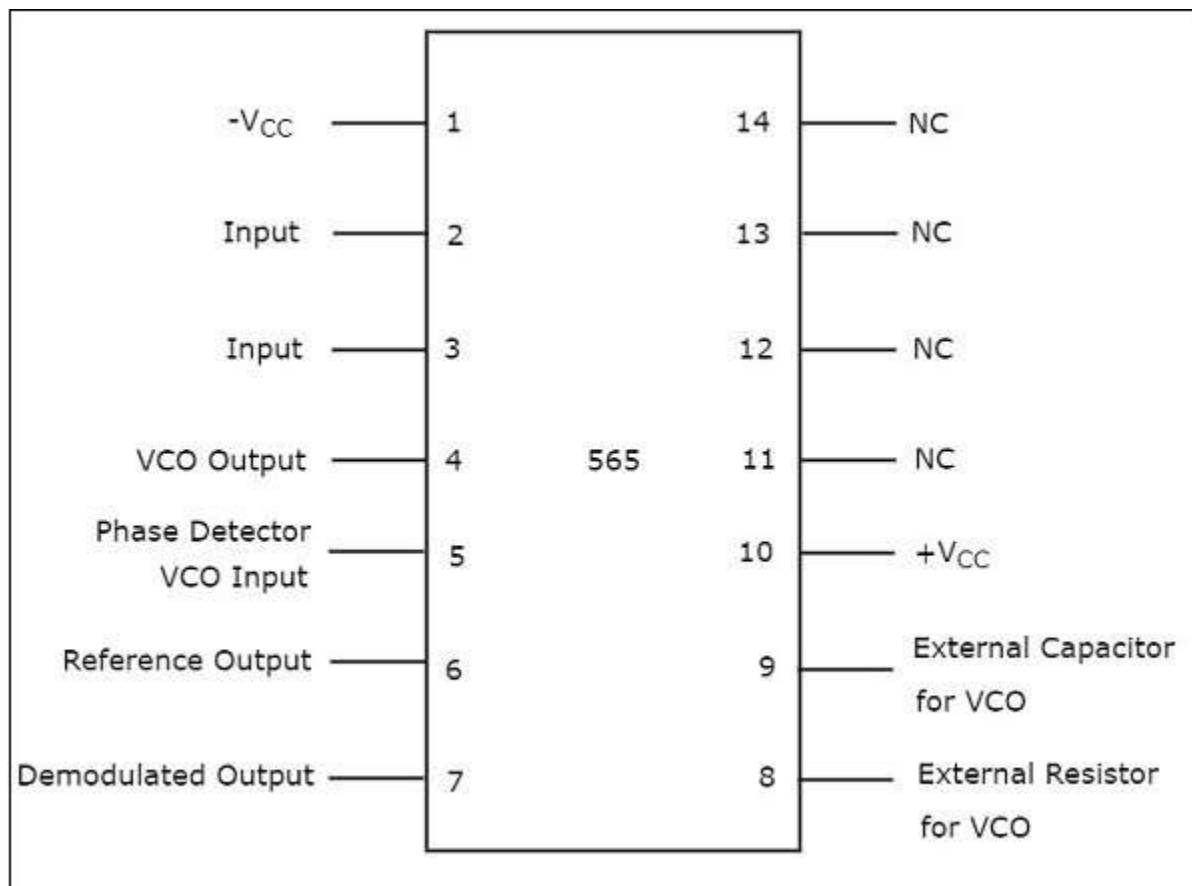
$$e(t) \propto \Delta \phi(t) \approx K_f m(t)$$

Thus, the **PLL output voltage** is proportional to the original message signal $m(t)$.

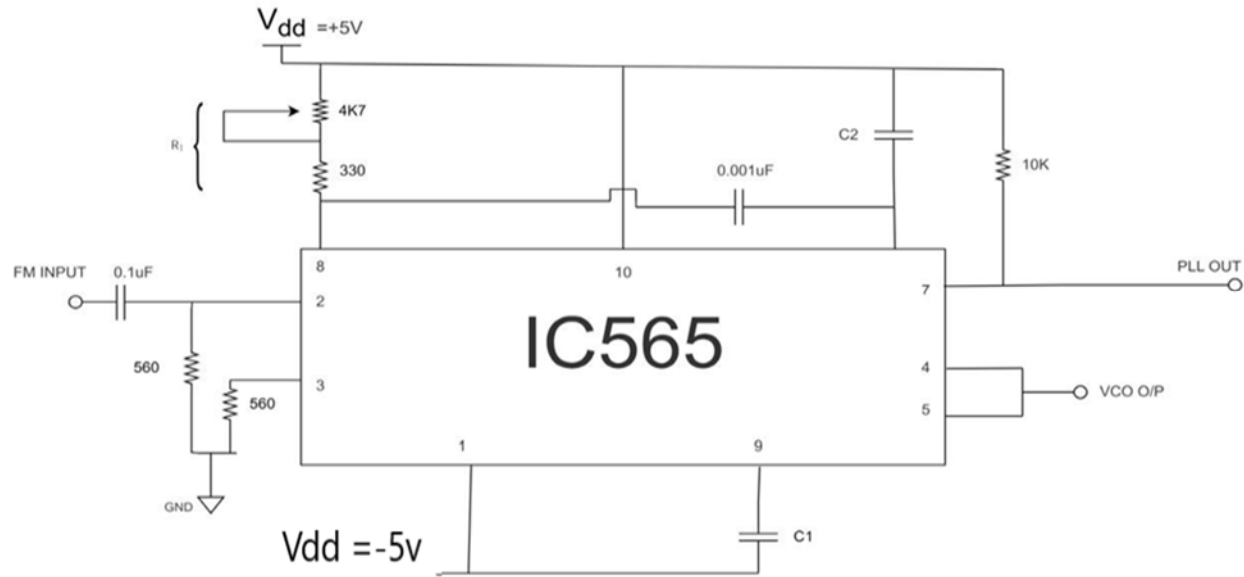
FM Demodulation based on PLL (Phase-Locked Loop):



IC 565 Pin diagram:



IC 565 Pin Connection with External Circuit



DESIGN:

Free running frequency of VCO,

Max= -12v to +12v Both I/P terminals (Pin 2 & 3) must receive identical DC bias range from 0 to 4V. R1 value is between 2KΩ to 20KΩ.

O/P Voltage swings –ve as I/P freq increases.

$$f_o = \frac{1}{4R_1C_1}$$

Filter (Internal):

Low pass filter

$$f = \frac{1}{2\pi \times 3.6 \times 10^3 \times C_2}$$

$$C_2 = 0.033 \mu F$$

$$f = \frac{1}{2 \times 3.141 \times 3.6 \times 10^3 \times .033 \times 10^{-6}}$$

$$= 1.3 \text{ KHz}$$

EXPECTED O/P WAVEFORMS:

1. Without providing the FM Input, check the internal VCO Output Waveform of PLL IC on the oscilloscope.
2. Adjust the potentiometer at R1, to set the frequency of VCO O/p to 10 KHz. (10 KHz is the carrier frequency chosen for modulation). (Waveform as shown in fig -1)
3. Provide the FM Input at pin 2 of PLL IC and observe the PLL O/p waveform at pin no 7 of IC. (waveform as shown in fig – 2).
4. An additional low pass filter can be added with the circuit to eliminate frequency components other than modulating signal frequency 1 KHz from PLL o/p.

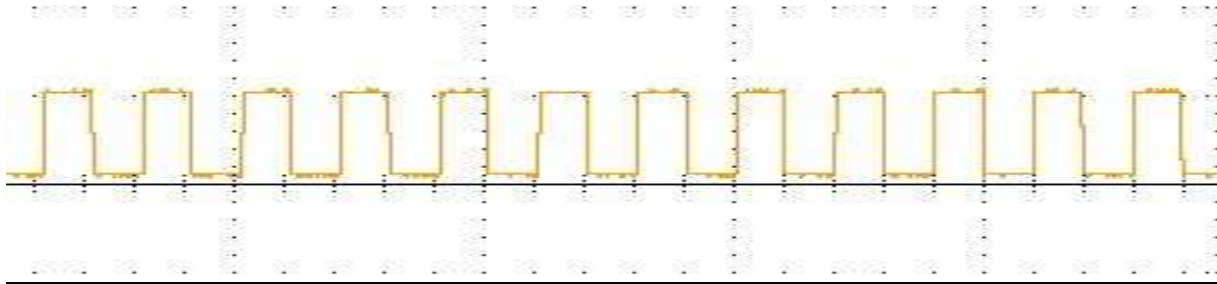


Fig -1. VCO Output Waveform of PLL IC

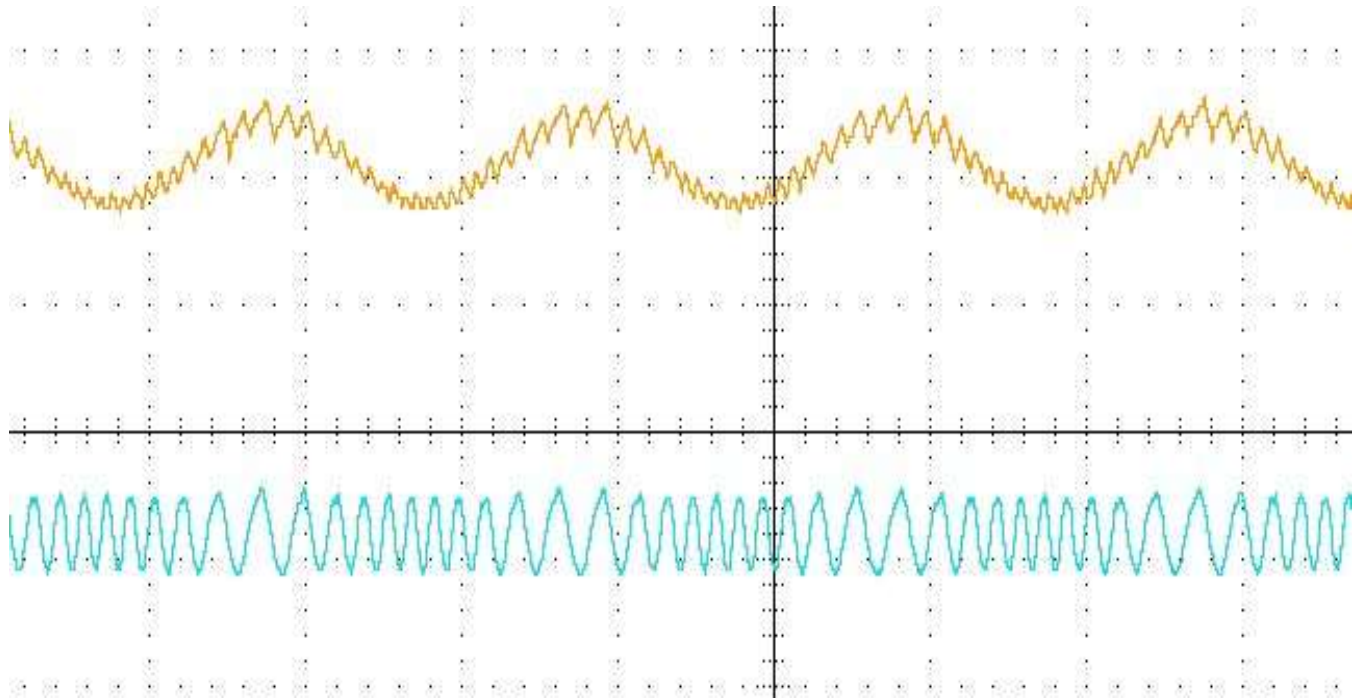


Fig -2. (a) PLL Output waveform (Demodulated signal) (b) FM Input Waveform